

## Lab 8 Logistic Regression

### Learning outcome

1. Describe the assumptions and process of regression analysis.
  2. Apply regression models on data variables.
  3. Interpret the outcomes of regression models for data-driven decisions.
  4. Recognize how data analytics is used in Smart Manufacturing environment.
- To apply logistic regression to predict categorical outcome
  - To predict Classification problem (categorical outcome), we use **Logistic Regression Learner** node.

Logistic Regression Learner



## Exercise 1: Classify iris species (*Data set : iris.csv*)

In this exercise, we want to **predict the species** of Iris flower, from petal length, width and sepal length, width. (4 independent variables).

The target, or dependent variable is the “Species”, which are Setosa, Versicolor and Virginica (3 possible outcomes).

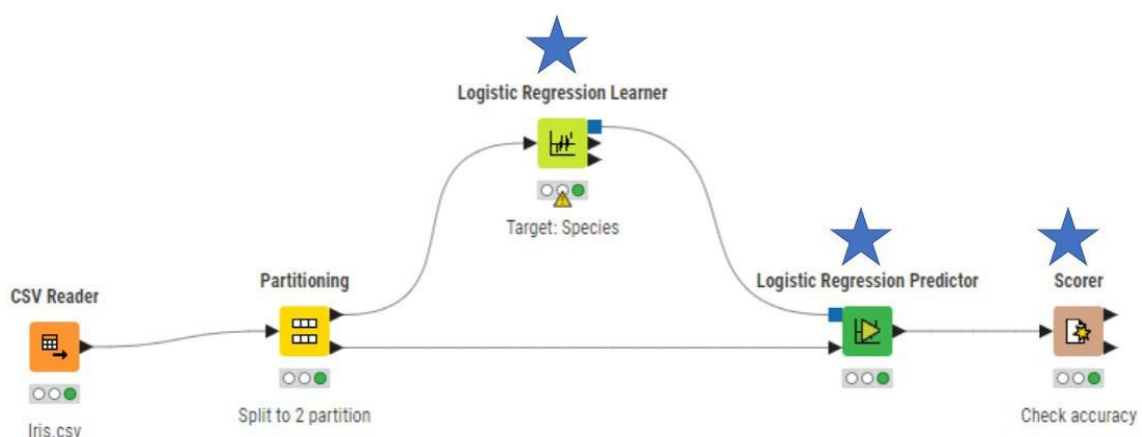
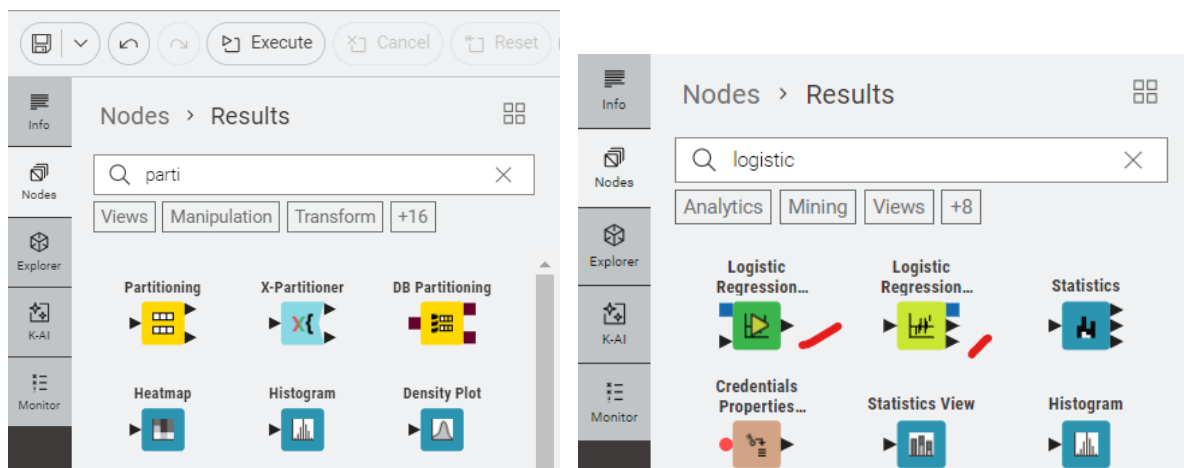
[\[Show me\]](#)

Launch KNIME.

Start a new Workflow. Save your workflow as “*Iris\_regression*”.

Setup your workflow with the following Nodes and connection:

\*Search for **Partitioning nodes, Logistic Regression Learner, Logistic Regression Predictor, Scorer**



Configure the **Partitioning node** : 70% first partition

Dialog Partitioning (Split to 2 partition) — □ ×

File

First partition Flow Variables Job Manager Selection Memory Policy

Choose size of first partition

☐ Absolute

☒ Relative[%]

☐ Take from top

☐ Linear sampling

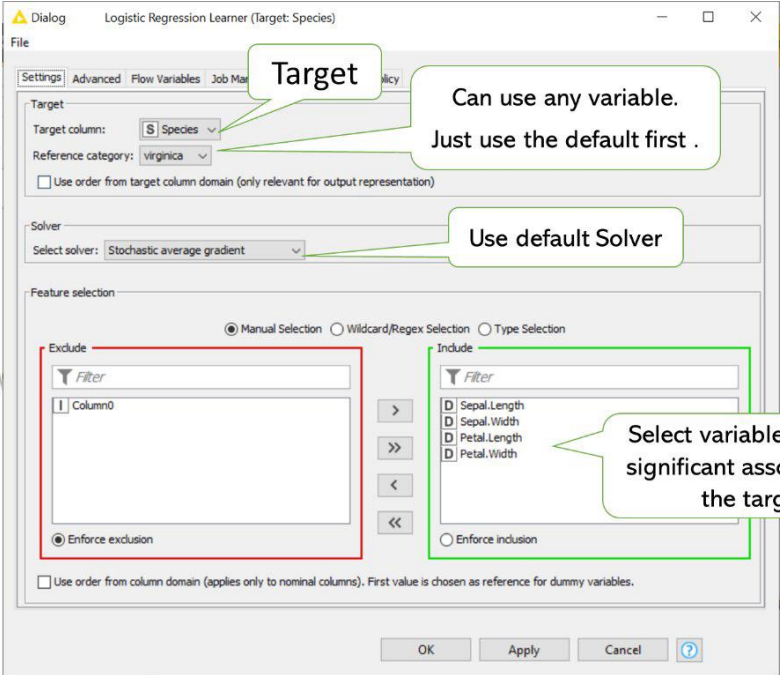
☒ Draw randomly

☐ Stratified sampling

☐ Use random seed

OK Apply Cancel ?

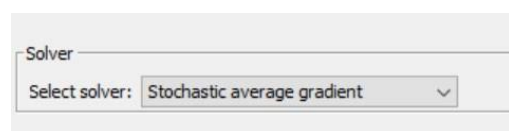
Configure the **Logistic Regression Learner** node:



The screenshot shows the 'Logistic Regression Learner (Target: Species)' dialog box. Annotations highlight key settings:

- Target:** Target column is set to 'Species'. Annotation: "Can use any variable. Just use the default first."
- Solver:** Select solver is set to 'Stochastic average gradient'. Annotation: "Use default Solver"
- Feature selection:** Manual Selection is chosen. The 'Include' list contains 'Sepal.Length', 'Sepal.Width', 'Petal.Length', and 'Petal.Width'. Annotation: "Select variables that has significant association to the target."

On the left, a small diagram shows the 'Logistic Regression Learner' node in a workflow, with 'Target: Species' indicated.



It dictates the algorithm use for the iteration.

This will not be covered in detail.

Which variable should you select to "Include" in the model?

There is no fast formula, but base on your exploratory analysis, you have seen Petal Length, Sepal Length, Petal Width and Sepal Width are able to differentiate the different species. Example:



No configuration needed for **Logistic Regression Predictor node**. Execute the node.  
A new column name "Prediction" will be automatically appended.

The screenshot shows an Orange3 workflow with the following nodes: CSV Reader, Partitioning (Split to 2 partition), Logistic Regression Learner (Target: Species), Logistic Regression Predictor, and Scorer. The workflow is executed, and the 'Predicted data' table is displayed below. The table has 7 columns: #, Row..., Column0, Sepal.Length, Sepal.Width, Petal.Length, and Petal.Width. The 'Species' and 'Prediction (Species)' columns are highlighted with a red dashed box.

| # | Row... | Column0 | Sepal.Length | Sepal.Width | Petal.Length | Petal.Width | Species | Prediction (Species) |
|---|--------|---------|--------------|-------------|--------------|-------------|---------|----------------------|
| 1 | Row4   | 5       | 5            | 3.6         | 1.4          | 0.2         | setosa  | setosa               |
| 2 | Row5   | 6       | 5.4          | 3.9         | 1.7          | 0.4         | setosa  | setosa               |
| 3 | Row8   | 9       | 4.4          | 2.9         | 1.4          | 0.2         | setosa  | setosa               |
| 4 | Row12  | 13      | 4.8          | 3           | 1.4          | 0.1         | setosa  | setosa               |
| 5 | Row13  | 14      | 4.3          | 3           | 1.1          | 0.1         | setosa  | setosa               |
| 6 | Row18  | 19      | 5.7          | 3.8         | 1.7          | 0.3         | setosa  | setosa               |
| 7 | Row20  | 21      | 5.4          | 3.4         | 1.7          | 0.2         | setosa  | setosa               |

Configure the **Scorer node**:

The screenshot shows the 'Scorer (Check accuracy)' dialog box. The 'First Column' is set to 'Species' and the 'Second Column' is set to 'Prediction (Species)'. The 'Sorting of values in tables' section shows 'Insertion order' selected. The 'Provide scores as flow variables' section has 'Use name prefix' unchecked. The 'Missing values' section has 'Ignore' selected.

**Scorer (Check accuracy)**

File

Scorer | Flow Variables | Job Manager Selection | Memory Policy

First Column:

Second Column:

Sorting of values in tables

Sorting strategy:  ☐ Reverse order

Provide scores as flow variables

☐ Use name prefix

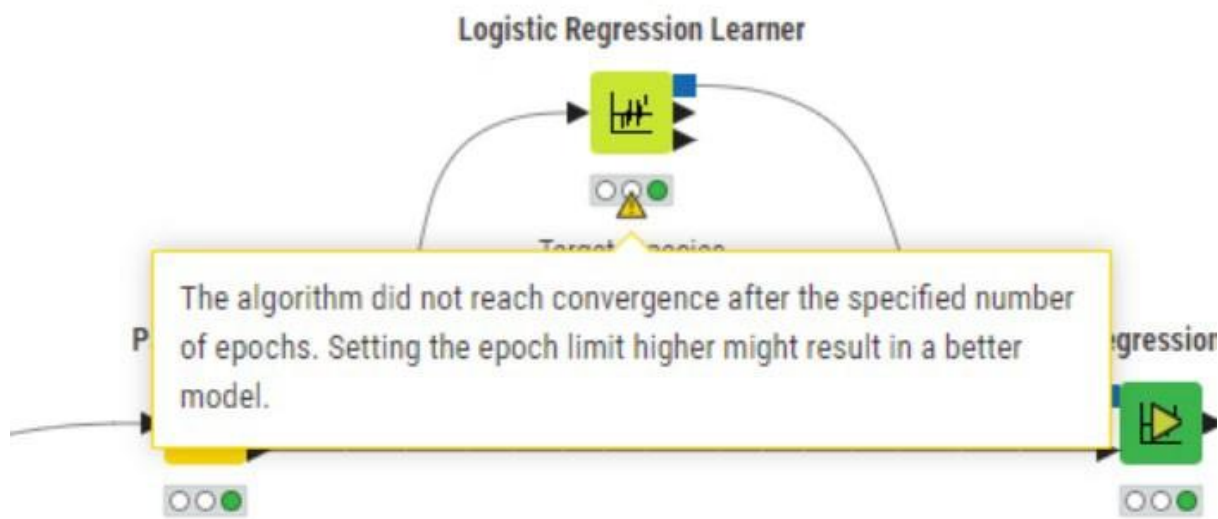
Missing values

In case of missing values: ☒ Ignore ☐ Fail

OK Apply Cancel ?

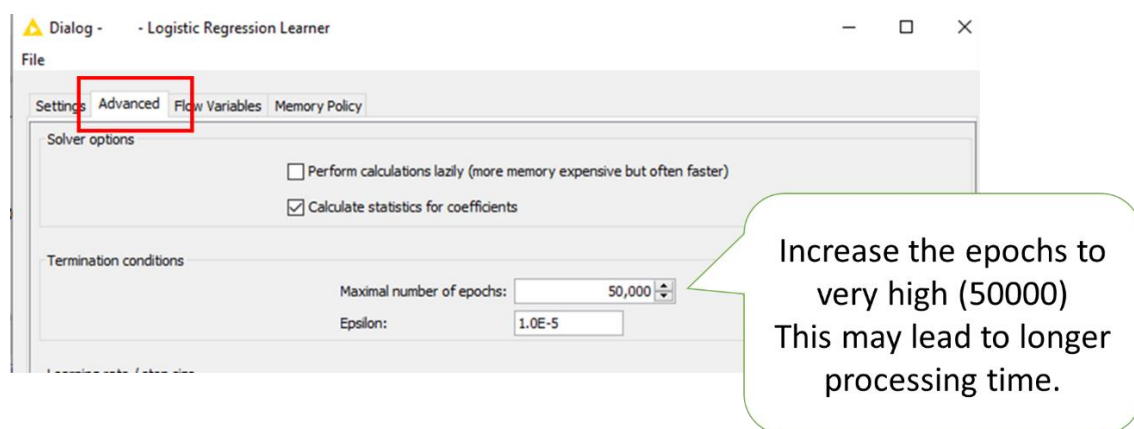
"Execute" the **Scorer node**.

Do the following if there is error message: Not able to converge



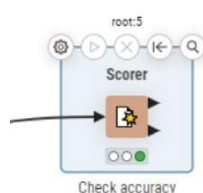
If there is error (not able to converge) in the **Logistic Regression Learner node**, edit the configuration:

(The model has been created. It is optional to increase the epochs number.)

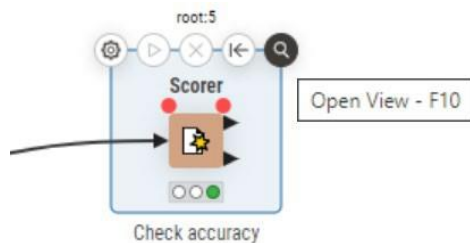


The number of epochs is a hyperparameter that defines the number of times that the learning algorithm will work through the entire training dataset.

**\*Note that logistic regression node is an Iterative process. Increasing the epochs number will result in longer processing time.**



Execute the **Scorer** node again



Open View – Open the Confusion Matrix



Confusion Matrix · Scorer (Check accuracy)

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□

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| Species \ P... | setosa | versicolor | virginica |
|----------------|--------|------------|-----------|
| setosa         | 17 ✓   | 0          | 0         |
| versicolor     | 0      | 12 ✓       | 0         |
| virginica      | 0      | 1          | 15 ✓      |

Correct classified: 44

Wrong classified: 1

Accuracy: 97.778%

Error: 2.222%

Cohen's kappa ( $\kappa$ ): 0.966%

Depend on your setting in **Partitioning node**, and **Logistic Regression Learner node**, you may or may not have the similar result as above.

*\*\*Although we are using the same set of data, but the content of 30% test set is different as they are randomly selected.*

The confusion matrix above, shows that 1 virginica species was wrongly classified as Versicolor.

With one error out of total 45 samples (30% of 150), the % accuracy is calculated as

$$\frac{44}{45} * 100\% = 97.8\%$$

As the model (algorithm) is a sigmoid curve equation, not like the linear equation which is simple equation, we will not study in this course.

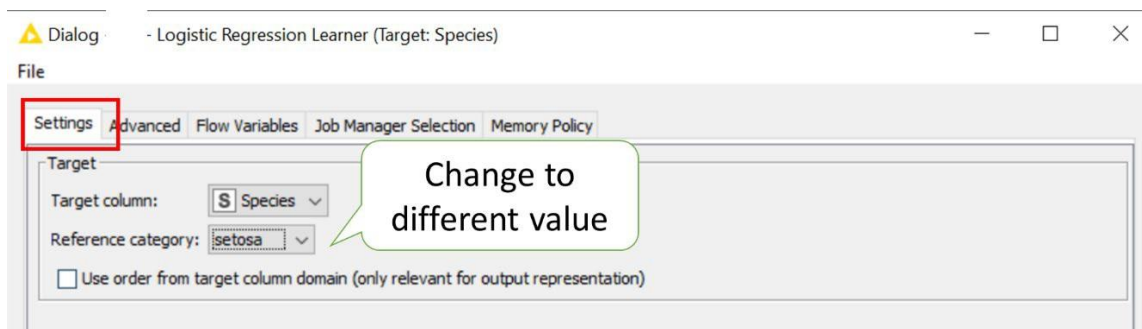
\***Cohen's kappa coefficient (κ)** is a [statistic](#) that is used to measure [inter-rater reliability](#) (and also [Intra-rater reliability](#)) for qualitative (categorical) items <sup>[1]</sup>. It is generally thought to be a more robust measure than simple percent agreement calculation, as κ takes into account the possibility of the agreement occurring by chance. (not tested in this course)

## Possible to improve the model to raise the accuracy?

Let's improve the accuracy by changing some parameters in "Logistic Regression Learner;"

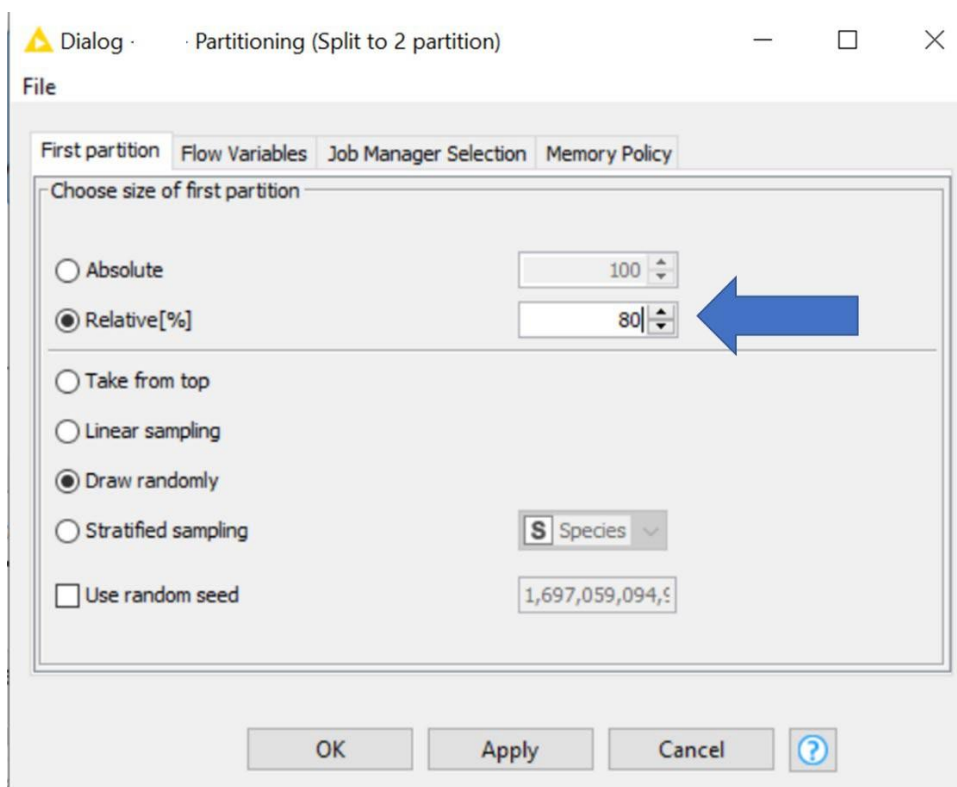
Let's try a different reference:

Re-configure the **Logistic Regression Learner**:



Execute the "**Scorer**" node to check the accuracy.

If the accuracy does not improve, edit the **partitioning** option:

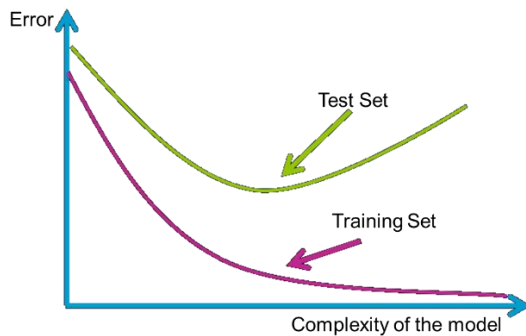


Execute the **Scorer** node again.



Not all machine model prediction can reach 100%. Depend on the behaviour of the data set and the algorithm setting. Usually above 80% are good enough. *Do not be over fixated to achieve 100% in the modelling result.*

Note that even if the training data gave 100% accurate, but when use with new test data, the accuracy may vary:



The training set may show high accuracy, but when use with new data, there may still be error. This is because the model has fitted too closely to the training set. This is called over fitting.

|  |                                                                                                                                                                                                                                                                                                                                                                                             |
|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <p>Your project data may not give you good accuracy.</p> <p>Reason for low accuracy could be:</p> <ul style="list-style-type: none"> <li>• Data set is not enough in total number</li> <li>• Data set does not cover all categories in equal number</li> <li>• Independent variables were not highly correlated with target</li> <li>• Logistic regression model is not suitable</li> </ul> |
|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

Remember to save your file.